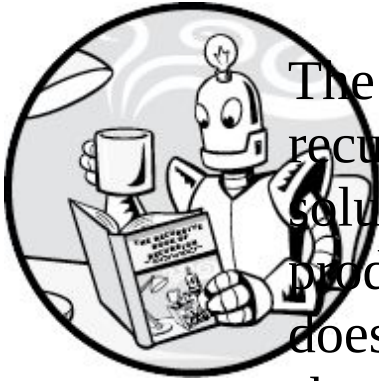


# INTRODUCTION



The programming technique of recursion can produce elegant code solutions. More often, however, it produces confused programmers. This doesn't mean programmers can (or should) ignore recursion. Despite its reputation for being challenging, recursion is an important computer science topic and can yield keen insights into programming itself. At the very least, knowing recursion can help you nail a coding job interview.

If you're a student with an interest in computer science, recursion is a necessary hurdle you'll have to overcome to understand many popular algorithms. If you're a programming bootcamp graduate or self-taught programmer who managed to bypass the more theoretical computer science topics, recursion problems are still sure to come up during whiteboard coding interviews. And if you're an experienced software engineer who has never touched a recursive algorithm before, you might find recursion to be an embarrassing gap in your knowledge.

There's no need to worry. Recursion isn't as hard to understand as it is to teach. As I'll explain in Chapter 1, I attribute the widespread misunderstanding of recursion to poor instruction rather than any inherent difficulty. And since recursive functions aren't commonly used in day-to-day programming, many folks get along just fine without them.